

Customer Churn Prediction in the Telecom Industry

A Comparative Evaluation of Decision Trees, Rule-Based Classification, k-Nearest Neighbours, Random Forest and AdaBoost

- **Course:** MDSDF405 – Machine Learning
- **Programme:** M.Sc. Data Science and Artificial Intelligence
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1. Objective, Business Context & Dataset

Customer churn is a major challenge for subscription-based telecom companies, since retaining an existing customer is generally cheaper than acquiring a new one. This project builds an end-to-end supervised learning pipeline to identify customers likely to churn before they leave, so the business can target them with retention offers. Five model families are trained, tuned and compared: Decision Tree, a rule-based classifier extracted from the tree, k-Nearest Neighbours (kNN), Random Forest and AdaBoost.

Dataset: [Telco Customer Churn](#) (Kaggle) – 7043 customers, 21 columns (20 features + binary target Churn). Each row represents one customer with demographic attributes, subscribed services, contract/billing details, and churn status. The class distribution is moderately imbalanced (~26.5% churn).

2. Task A – Data Understanding & Preprocessing

2.1 Data Quality

No duplicate records were found. During type correction, TotalCharges (stored as text) revealed 11 hidden missing values, all belonging to customers with tenure = 0, meaning newly acquired customers with no billing history yet. Rather than imputing with mean/median (which would fabricate a billing history that does not exist), these were filled with 0. Numerical features (tenure, MonthlyCharges, TotalCharges) were screened for outliers using boxplots and the IQR method; flagged points corresponded to genuine long-tenure or premium-plan customers rather than data errors, so no observations were removed.

2.2 Attribute Classification (non-obvious cases)

Attribute	Type	Reasoning
SeniorCitizen	Nominal (binary)	Stored as 0/1 integer, but represents category membership, not a quantity or order.
Contract	Ordinal	Month-to-month / One year / Two years have a natural duration ordering, unlike other service flags.
TotalCharges	Numeric (continuous)	Stored as text due to blank entries; storage type and true measurement scale differ.

2.3 Encoding, Scaling & Split

- Binary Yes/No fields and gender were label-encoded (0/1)
- Contract was ordinally encoded using its natural duration order
- The remaining nominal service/payment fields were one-hot encoded

The dataset was split **80:20 (train:test)** using **stratified sampling** on Churn before one-hot encoding and scaling were fitted, and the fitted transformers were then applied unchanged to the test set. This leakage-free ordering ensures the test metrics reported below are an unbiased estimate of generalisation. Stratification was chosen because of the ~27% class imbalance, so both partitions preserve the true churn rate (26.5% in

each). Continuous features (tenure, MonthlyCharges, TotalCharges) were standardised with **StandardScaler** (mean 0, std 1) for the distance-based kNN model. Tree-based models used the unscaled features since they split on thresholds, not distances.

3. Task B – Decision Tree: Training, Tuning & Evaluation

A reusable evaluation function computes the Confusion Matrix, Accuracy, Precision, Recall (Sensitivity), Specificity and Negative Predictive Value (NPV) for every model.

Metric	Baseline Decision Tree	Tuned Decision Tree	Δ Change
Accuracy	0.737	0.798	+0.061
Precision	0.504	0.662	+0.158
Recall	0.497	0.487	-0.010
Specificity	0.823	0.910	+0.087
NPV	0.819	0.831	+0.012

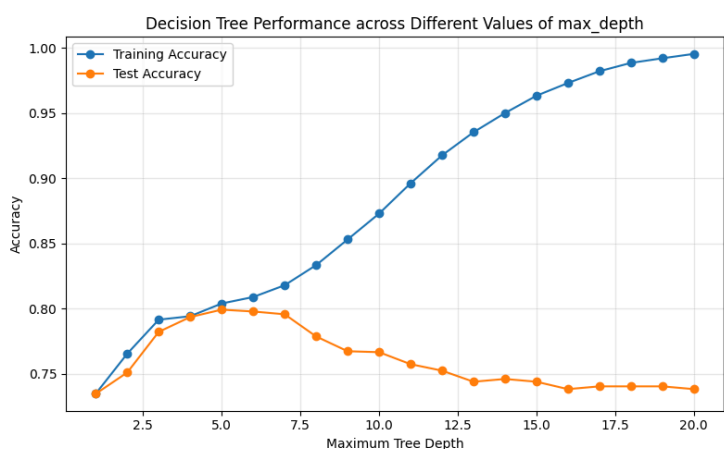


Fig. 1: Train vs. test accuracy

The unconstrained baseline tree reached depth 24 with 1088 leaves, scoring 99.8% training accuracy but only 73.7% test accuracy – a classic case of high-variance overfitting. Diagnostic sweeps varied *max_depth*, *min_samples_split* and *min_samples_leaf* one at a time (others held at their defaults).

As *max_depth* increases, training accuracy rises monotonically toward 100% while test accuracy peaks around depth 5-6 then declines (Fig. 1) – the classic overfitting curve.

min_samples_split and *min_samples_leaf* show the mirror-image pattern (Fig. 2): both start at their most flexible setting (near-maximum train/test accuracy) and, as the constraint is tightened, force increasingly simple, high-bias trees, so both train and test accuracy fall together.

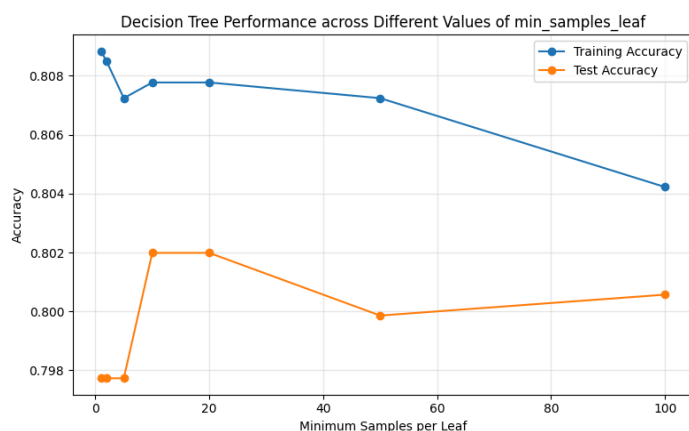
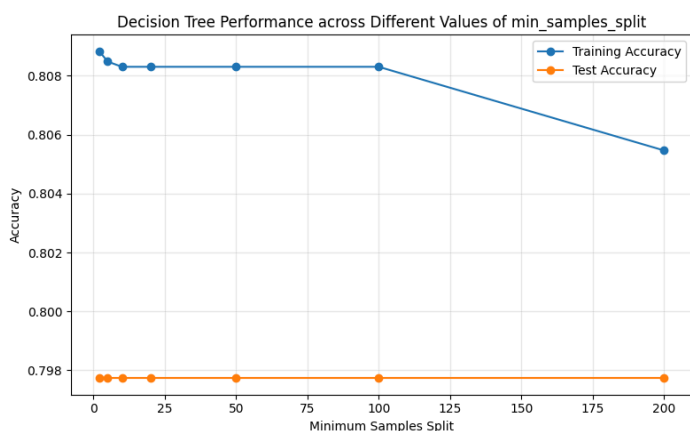


Fig. 2: Train vs. test accuracy as *min_samples_split* (left) and *min_samples_leaf* (right) increase – both fall together as the constraint tightens, the opposite direction to *max_depth*, since larger values force simpler, higher-bias trees.

All three curves illustrate the bias-variance trade-off, just approached from opposite directions. Five-fold stratified cross-validation on the training set alone selected *max_depth* = 6, *min_samples_split* = 50, *min_samples_leaf* = 1 (mean CV accuracy 79.1%), improving test accuracy to 79.8% and precision to 66.2%, at a negligible recall cost.

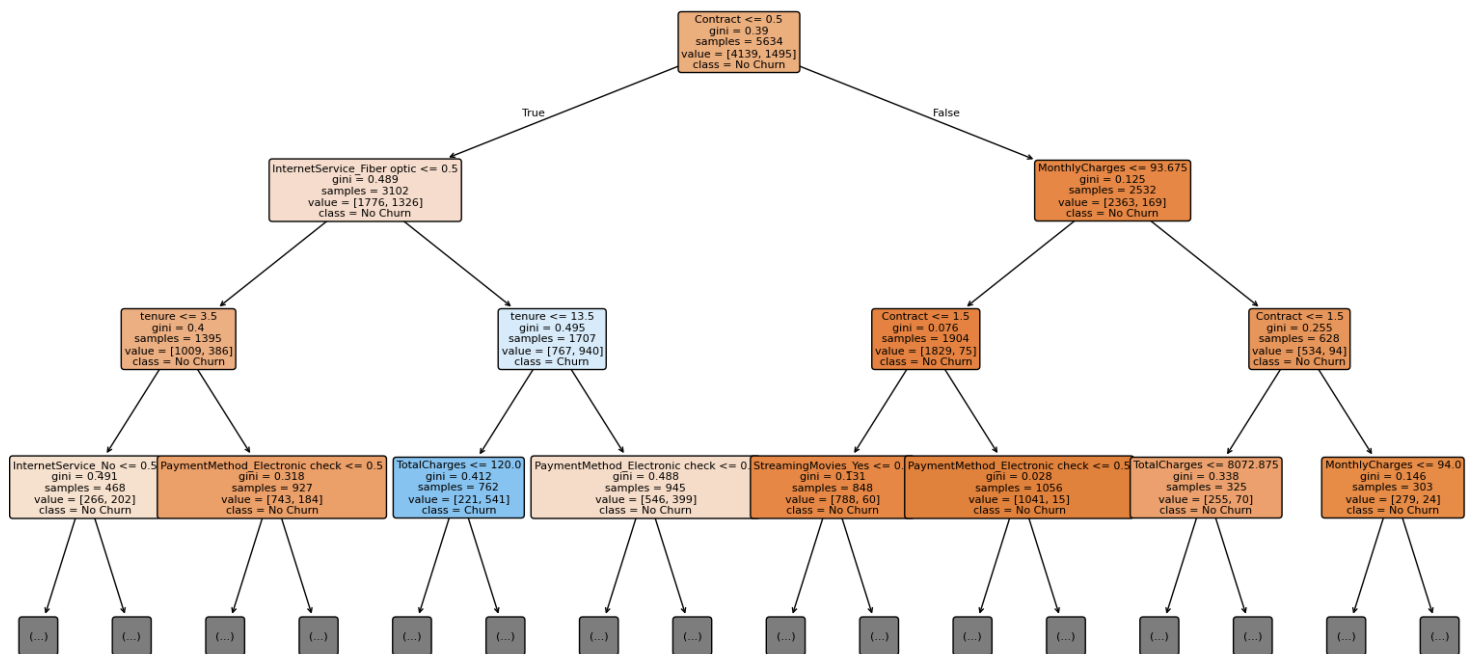


Fig. 3: Top three levels of the tuned Decision Tree. The root split on Contract confirms it as the single strongest churn predictor.

Interpretation of Two Decision Paths

Higher-risk path: Month-to-month contract → Fibre-optic internet → short tenure → high Monthly/Total Charges → predicted Churn. Recently acquired customers on expensive, flexible contracts are the model's highest-risk profile.

Lower-risk path: One-year/Two-year contract → split by Monthly Charges, Payment Method, Senior Citizen status → predominantly predicted No Churn. Long-term contractual commitment is the single strongest retention signal, consistent with Contract being the root-node split.

4. Task C – Rule-Based Classification

Traversing the tuned tree yields 53 root-to-leaf IF-THEN rules.

The five highest-support rules below (with Contract ordinally encoded 0=Month-to-month, 1=One year, 2=Two year) all condition on **Contract**, confirming it as the dominant predictor. Four of the five predict No Churn with 82.6–100% leaf confidence, reflecting class imbalance, while the sole Churn-predicting rule combines month-to-month contracts with fibre-optic service and specific billing thresholds.

- **Rule 1** (support=624, confidence=0.987): IF Contract > 0.50 AND MonthlyCharges ≤ 93.67 AND Contract > 1.50 AND PaymentMethod_Electronic check ≤ 0.50 AND SeniorCitizen ≤ 0.50 AND OnlineBackup_Yes ≤ 0.50 → **THEN No Churn**
- **Rule 2** (support=375, confidence=0.979): IF Contract > 0.50 AND MonthlyCharges ≤ 93.67 AND Contract ≤ 1.50 AND StreamingMovies_Yes ≤ 0.50 AND PaymentMethod_Electronic check ≤ 0.50 AND OnlineBackup_Yes ≤ 0.50 → **THEN No Churn**
- **Rule 3** (support=304, confidence=0.826): IF Contract ≤ 0.50 AND InternetService_Fiber optic ≤ 0.50 AND tenure > 3.50 AND PaymentMethod_Electronic check ≤ 0.50 AND PhoneService > 0.50 AND tenure ≤ 17.50 → **THEN No Churn**

- **Rule 4** (support=298, confidence=1.000): IF Contract > 0.50 AND MonthlyCharges ≤ 93.67 AND Contract > 1.50 AND PaymentMethod_Electronic check ≤ 0.50 AND SeniorCitizen ≤ 0.50 AND OnlineBackup_Yes > 0.50 → **THEN No Churn**
- **Rule 5** (support=272, confidence=0.555): IF Contract ≤ 0.50 AND InternetService_Fiber optic > 0.50 AND tenure > 13.50 AND PaymentMethod_Electronic check > 0.50 AND TotalCharges ≤ 3078.40 AND TotalCharges ≤ 3001.07 → **THEN Churn**

Because the rules were extracted directly from the tuned tree (Option 1), the rule set reproduces the tree's decisions exactly (accuracy 79.8%, identical confusion matrix), so no interpretability-vs-accuracy trade-off is observed here. A dedicated induction algorithm such as RIPPER would typically learn a smaller, more compact rule set at a modest cost in accuracy – the classical trade-off would appear in that case, but not when rules are simply read off an existing tree.

5. Task D – kNN (Lazy Learning)

kNN was evaluated at $k = 1, 3, 5, 7, 9$ on scaled features. Test accuracy rose from 74.0% ($k=1$) to a plateau of 77.1% at $k=7-9$ (Fig. 4). 5-fold cross-validation confirmed **$k=9$ as optimal** (final test accuracy 77.1%, precision 56.8%, recall 56.7%, specificity 84.4%).

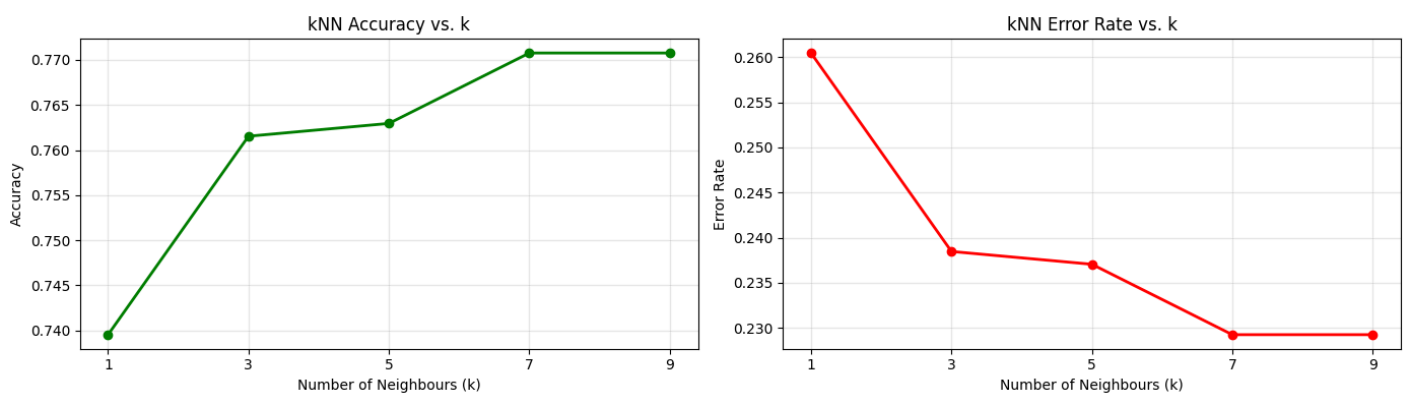


Fig. 4: kNN test accuracy and error rate across $k = 1, 3, 5, 7, 9$.

Why scaling matters:

kNN classifies by Euclidean distance, so unscaled features with large ranges (TotalCharges up to ~8000; tenure 0–72) would dominate the distance metric over 0/1-valued one-hot features. Run empirically at $k=9$, scaling gave 78.8% vs. 77.1% unscaled – a modest 1.7-point gap, smaller than typically expected, because most predictors in this dataset are already binary/one-hot and only three features have divergent scales. The direction confirms scaling helps; its small magnitude here is dataset-specific, not evidence that scaling is unimportant in general.

Bias-variance:

$k=1$ gives a highly flexible, low-bias/high-variance boundary sensitive to noise. Increasing k smooths the boundary (higher bias, lower variance) and improves generalisation, until k becomes so large the model oversmooths and underfits – consistent with the accuracy curve plateauing rather than continuing to climb.

6. Task E – Ensemble Learning

Random Forest (bagging) and AdaBoost (boosting) were each trained at default settings, then tuned via 5-fold GridSearchCV.

Random Forest: best params `n_estimators=100`, `max_depth=10`, `max_features='sqrt'` (mean CV accuracy 80.5%); test accuracy improved 78.8% → 80.3%, precision 62.8% → 66.7%, recall 49.2% → 51.9%.

AdaBoost: best params `n_estimators=200`, `learning_rate=0.5` (mean CV accuracy 80.3%); test accuracy improved 78.8% → 79.3%, recall 51.9% → 52.1%, precision 62.0% → 63.5%.

Random Forest Feature Importance:

The tuned Random Forest's feature importances (Fig. 6) confirm the same drivers identified by the Decision Tree: billing and contractual behaviour - TotalCharges, tenure, Contract, and MonthlyCharges - dominate, well ahead of service-related features like internet type or payment method. Demographic attributes rank lowest.

This agreement between two structurally different models- a single tree and 100-tree ensemble, is stronger evidence of the true churn drivers than either model's ranking alone.

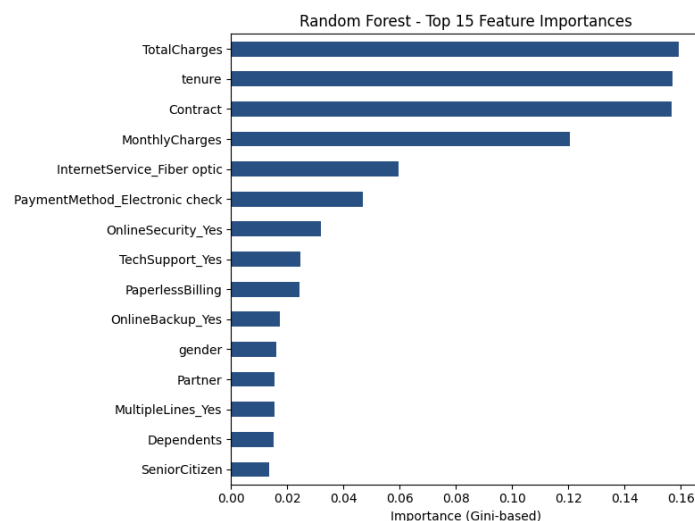


Fig. 5: Top 15 Feature Importance - tuned Random Forest

Note: TotalCharges is mechanically tied to tenure × MonthlyCharges, so the three are best read as one combined "billing history" signal rather than independent drivers.

Why ensembles beat a single tree:

A single deep tree has high variance – small data changes reshape it substantially.

Random Forest averages many de-correlated trees (bootstrap samples + random feature subsets at each split), which reduces variance.

AdaBoost instead trains weak learners sequentially, reweighting misclassified instances each round, which primarily reduces bias by progressively capturing harder decision boundaries.

Both mechanisms outperformed the single Decision Tree here, with the tuned Random Forest achieving the best overall accuracy of the two.

7. Task F – Final Comparison & Recommendation

Model	Accuracy	Precision	Recall	Specificity	Notes
Decision Tree (Baseline)	0.737	0.504	0.497	0.823	High variance; severe overfitting
Decision Tree (Tuned)	0.798	0.662	0.487	0.910	Reduced variance via CV tuning
Rule-Based	0.798	0.662	0.487	0.910	Identical to Tuned DT; fully interpretable
kNN (k=9)	0.771	0.568	0.567	0.844	Distance-based; k chosen via CV
Random Forest (Baseline)	0.788	0.628	0.492	0.895	Baseline Random Forest model
Random Forest (Tuned)	0.803	0.667	0.519	0.906	Best overall; bagging reduces variance
AdaBoost (Baseline)	0.788	0.620	0.519	0.885	Baseline AdaBoost model
AdaBoost (Tuned)	0.793	0.635	0.521	0.892	Sequential boosting; reduces bias

Confusion Matrix – Rule-Based (Tree-Derived Rules)

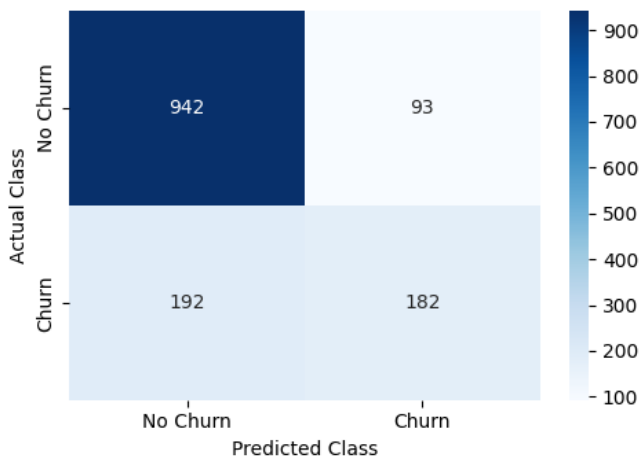


Fig. 6: Confusion matrix of the Rule-Based (Tree-Derived) classifier.

Since the rules were extracted directly from the tuned Decision Tree, the confusion matrix is identical to that of the tuned Decision Tree.

ROC Curve Comparison Across Models

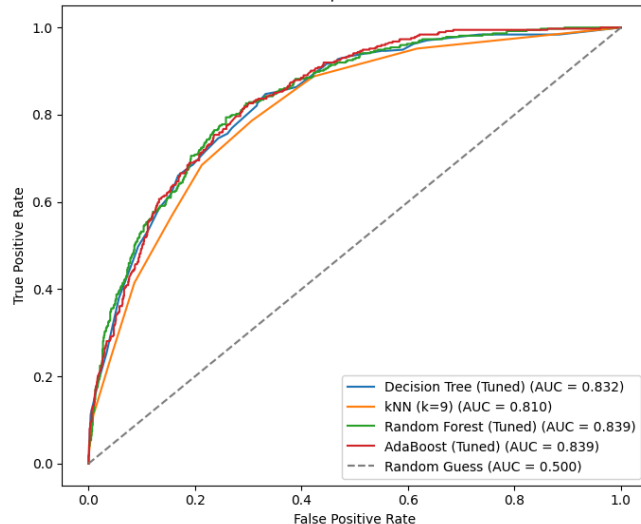


Fig. 7: ROC Curve Comparison Across Models

Best Model & Why

The **tuned Random Forest** achieves the best overall accuracy (0.803) and precision (0.667), consistent with the bias-variance trade-off: bagging averages many overfit, high-variance trees into a lower-variance ensemble without raising bias. kNN attains the highest recall (0.567), while the tuned Decision Tree/rule set achieves the highest specificity (0.910). No single model dominates on every metric, so the right choice depends on which error type the business weighs most heavily. Notably, on ranking ability (AUC) AdaBoost is statistically tied with Random Forest (0.839 each), so the accuracy gap between them reflects where each model's default threshold sits rather than a real difference in underlying discriminative power- a distinction that matters if the deployed threshold is later retuned.

Interpretability

The **tuned Decision Tree and its extracted rule set** are the most interpretable: a retention agent can see exactly why a customer was flagged (e.g. month-to-month contract, low tenure, fibre-optic service) and choose a targeted intervention. An opaque ensemble score cannot explain itself as directly, even though Random Forest predicts slightly better. In a churn-management context, this transparency is a practical requirement for stakeholder trust and actionability, not a mere convenience.

Deployment risks

False negatives (missed churners) are the costliest error – lost recurring revenue with no retention attempt. False positives waste retention budget on loyal customers. Data drift from evolving pricing, competition and customer behaviour will degrade performance over time, requiring monitoring and periodic retraining.

Fairness is also a concern: features such as gender and SeniorCitizen could introduce demographic bias into retention targeting and should be audited across customer subgroups before deployment.

Conclusion

An end-to-end, leakage-free pipeline was built and evaluated across five model families. Hyperparameters were selected via cross-validation on the training set only, with the test set reserved for a single unbiased final evaluation. If the priority is **maximum predictive performance**, the **tuned Random Forest** is the strongest candidate. If the priority is **transparency and actionability**, the **tuned Decision Tree** or its rule-based representation is the more defensible choice. There is no single universally best model – the right choice depends on the business objective.